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Mathématiques et de statistique
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Mathematics and Statistics

Introduction to linear algebra MAT 1341 C
Test 2 – version A

18 March, 2025. 13:00

Instructions:

- The length of this test is **75 minutes**.
- Questions 1 and 2 consist of short-answer questions. You do not need to justify your answers in this question.
- Questions 3 to 5 are long-answer questions. **To get full marks on these questions, you must justify your answers, and your solutions must be clear, logical, and readable.**
- Question 6 is a bonus question with 3 points. To get marks on this question, your solution must be completely correct. You should not work on this question before you have finished all the others.
- There is a blank page at the end of the test. If you need extra space for any question, please use the blank page and/or the backs of pages. Indicate clearly where your answer can be found.
- **Cellular phones, smart watches, unauthorized electronic devices, or course notes are not allowed during this exam. Phones and devices must be turned off and put away in your bag. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.**

  By signing below, you acknowledge that you have ensured that you are complying with the above statement.
- Good luck!

Last name :_____ First name :_____

Student number: _____ Signature :_____

Q1	Q2	Q3	Q4	Q5	B	Total
6	4	10	10	10	3	40

(1) Determine whether each of the following statements is true or false.

(a) If $A \in M_{n\times n}(\mathbb{R})$ and $v \in \mathbb{R}^n$ then $\{Av, A^2v, \ldots, A^{n+1}v\}$ is always linearly dependent. Solution: true

(b) If $\{v_1, \ldots, v_7\}$ is a LI set in a vector space V then $\{v_1, v_2, v_3\}$ must also be LI. Solution: true

(c) If $A, B \in M_{n\times n}(\mathbb{R})$, and B is the inverse of A, then $AB = BA$. Solution: true

(d) Given a matrix $A \in M_{m\times n}(\mathbb{R})$, if $\tilde{A}$ is the RREF of A, then $\mathrm{Row}(A) = \mathrm{Row}(\tilde{A})$. Solution: true

(e) If $\{v_1, \ldots, v_{16}\}$ is a LI set in a vector space V and $\{w_1, \ldots, w_{16}\}$ is a spanning set, then $\{v_1, \ldots, v_{16}\}$ must be a basis. Solution: true

(f) For any non-zero matrices $A, B \in M_{n\times n}(\mathbb{R})$, if AB is invertible, then A is invertible. Solution: true

Below, circle T for statements that are true and F for statements that are false. [6]

(a)	(b)	(c)	(d)	(e)	(f)
T/F	T/F	T/F	T/F	T/F	T/F

(2) Let A be a 13×22 matrix such that $\mathrm{rank}(A) = 10$. Answer the following questions. You do not need to justify your answers.

(a) If the column space of A^T is a subspace of $\mathbb{R}^n$, what is the value of n? _____ [1]

Solution: 22

(b) What is the dimension of the null space of A^T? _____ [1]

Solution: 3

(c) What is the dimension of the null space of A? _____ [1]

Solution: 12

(d) What is the dimension of the row space of A? _____ [1]

Solution: 10

(3) (a) Determine whether the set $\left\{ \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -2 \\ -1 & 0 \end{bmatrix} \right\}$ is linearly independent or dependent. [5]

Solution: Suppose $a, b, c \in \mathbb{R}$ satisfy

$$a \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} + c \begin{bmatrix} 1 & -2 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

This gives

$$\begin{bmatrix} a+c & b-2c \\ a-c & 2a+b \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

which gives the equations

$$\begin{aligned} (1) \quad & a + c = 0 \\ (2) \quad & b - 2c = 0 \\ (3) \quad & a - c = 0 \\ (4) \quad & 2a + b = 0. \end{aligned}$$

We solve this:

$$\begin{aligned} (1)+(3) : 2a = 0 \quad & \therefore \ a = 0 \\ (1) : 0 + c = 0 \quad & \therefore c = 0 \\ (2) : b - 2(0) = 0 \quad & \therefore \ b = 0. \end{aligned}$$

Therefore they are LI.

Alternative solution. Identify these vectors with their coordinates in the basis

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

of $M_2(\mathbb{R})$, to get the vectors $(1, 0, 1, 2), (0, 1, 0, 1), (1, -2, -1, 0)$.

Next put these as rows in a matrix, and row-reduce.

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 1 & -2 & -1 & 0 \end{bmatrix} \quad R_3 - R_1 \to \quad \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & -2 & -2 & -2 \end{bmatrix}$$

$$R_3 + 2R_2 \to \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & 0 \end{bmatrix} \quad -(1/2)R_3 \to \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The REF has a leading one in every row, which means that the original matrices were linearly independent.

(b) Extend the basis $\{(1,2,3,4),(2,4,7,8)\}$ to a basis in $\mathbb{R}^3$ $\mathbb{R}^4$. [5]

Solution: Use these vectors as the rows of a matrix, and then use the Row Space Algorithm:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 7 & 8 \end{bmatrix} \qquad R_2 - 2R_1 \to \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

We have no leading ones in columns 2 and 4. Therefore adding the vectors e_2 and e_4 will produce a linearly independent set.

Hence a basis is:

$$\{(1,2,3,4),(2,4,7,8),(0,1,0,0),(0,0,0,1)\}.$$

(4) Let $A = \begin{bmatrix} 1 & 3 & 2 & 4 \\ 2 & 6 & 5 & 8 \\ 1 & 3 & 1 & 2 \end{bmatrix}$.

(a) Find a basis of the column space of A. [5]

Solution: Use the Column Space Algorithm.

$$\begin{bmatrix} 1 & 3 & 2 & 4 \\ 2 & 6 & 5 & 8 \\ 1 & 3 & 1 & 2 \end{bmatrix} \quad \begin{matrix} R_2 - 2R_1 \to \\ R_3 - R_1 \to \end{matrix} \quad \begin{bmatrix} 1 & 3 & 2 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & -2 \end{bmatrix}$$

$$R_2 + R_3 \to \begin{bmatrix} 1 & 3 & 2 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} \quad -(1/2)R_3 \to \begin{bmatrix} 1 & 3 & 2 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

There are leading ones in columns 1, 3, and 4; therefore the corresponding columns from A form a basis. A basis for $\mathrm{Col}(A)$ is $\{(1,2,1),(2,5,1),(4,8,2)\}$.

(b) Find a basis of the ~~column~~ row space of A. [5]

Solution: Use the Row Space Algorithm. We found a REF in (a); therefore the non-zero rows form a basis.

A basis for $\mathrm{Row}(A)$ is $\{(1,3,2,4),(0,0,1,0),(0,0,0,1)\}$.

(5) (a) Find the inverse of the matrix $\begin{bmatrix} 1 & 6 & 3 \\ 1 & 8 & 4 \\ 2 & 3 & 2 \end{bmatrix}$. [5]

Solution: We start with the matrix $\left[\begin{array}{ccc|ccc} 1 & 6 & 3 & 1 & 0 & 0 \\ 1 & 8 & 4 & 0 & 1 & 0 \\ 2 & 3 & 2 & 0 & 0 & 1 \end{array}\right]$ and find its RREF.

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 3 & 1 & 0 & 0 \\ 1 & 8 & 4 & 0 & 1 & 0 \\ 2 & 3 & 2 & 0 & 0 & 1 \end{array}\right] \quad \begin{matrix} R_2 - R_1 \to \\ R_3 - 2R_1 \to \end{matrix} \quad \left[\begin{array}{ccc|ccc} 1 & 6 & 3 & 1 & 0 & 0 \\ 0 & 2 & 1 & -1 & 1 & 0 \\ 0 & -9 & -4 & -2 & 0 & 1 \end{array}\right]$$

$$R_3 + 5R_2 \to \left[\begin{array}{ccc|ccc} 1 & 6 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & -7 & 5 & 1 \\ 0 & -9 & -4 & -2 & 0 & 1 \end{array}\right] \quad \begin{matrix} R_1 - 6R_2 \to \\ R_3 + 9R_2 \to \end{matrix} \quad \left[\begin{array}{ccc|ccc} 1 & 0 & -3 & 43 & -30 & -6 \\ 0 & 1 & 1 & -7 & 5 & 1 \\ 0 & 0 & 5 & -65 & 45 & 10 \end{array}\right]$$

$$\tfrac{1}{5}R_3 \to \left[\begin{array}{ccc|ccc} 1 & 0 & -3 & 43 & -30 & -6 \\ 0 & 1 & 1 & -7 & 5 & 1 \\ 0 & 0 & 1 & -13 & 9 & 2 \end{array}\right] \quad \begin{matrix} R_1 + 3R_3 \to \\ R_2 - R_3 \to \end{matrix} \quad \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 4 & -3 & 0 \\ 0 & 1 & 0 & 6 & -4 & -1 \\ 0 & 0 & 1 & -13 & 9 & 2 \end{array}\right].$$

Therefore the inverse is $\begin{bmatrix} 4 & -3 & 0 \\ 6 & -4 & -1 \\ -13 & 9 & 2 \end{bmatrix}$.

(b) Use your answer to part (a) to solve the following linear system.

$$\begin{cases} x + 6y + 3z = 5 \\ x + 8y + 4z = 6 \\ 2x + 3y + 2z = 5 \end{cases}$$

[5]

Solution:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 4 & -3 & 0 \\ 0 & 1 & 0 & 6 & -4 & -1 \\ 0 & 0 & 1 & -13 & 9 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Bonus: Let V be a vector space. Suppose that U and W are subspaces of dimension such that $\dim(U) = \dim(W) = 2$ and $U \cap W\{\vec{0}\}$. Prove that $\dim(V) \geq 4$. [3]

Solution: Let $\{u_1, u_2\}$ be a basis for U and $\{w_1, w_2\}$ be a basis for W. Let us show that $\{u_1, u_2, w_1, w_2\}$ is LI:

If $a_1u_1 + a_2u_2 + b_1w_1 + b_2w_2 = \vec{0}$ then

$$a_1u_1 + a_2u_2 = -b_1w_1 - b_2w_2.$$

The LHS is in U and the RHS is in W, so both sides are in $U \cap W = \{\vec{0}\}$.

Hence $a_1u_1 + a_2u_2 = \vec{0}$ so $a_1 = a_2 = 0$ since $\{u_1, u_2\}$ are LI. Likewise $b_1 = b_2 = 0$.

Therefore there is an LI set of size 4 in V, so $\dim(V) \geq 4$.